INFO 7375 · GRADUATE SEMINAR · SPRING 2026

NORTHEASTERN UNIVERSITY · KHOURY COLLEGE

Nik Bear Brown, PhD MBA
Associate Teaching Professor
ni.brown@neu.edu · Office 505A Dana Hall

Computational *Skepticism* for AI

A graduate course in the methodology of AI validation — the systematic practice of doubting, testing, and documenting the limits of AI systems before and during deployment.

CREDITS	FORMAT	STRUCTURE	OUTCOME
4	In-person, Boston	6 parts · 14 chapters	A validation pipeline you can defend

THE COURSE, IN ONE PARAGRAPH

There is a gap between the ability to *build and deploy* a model — and the ability to *evaluate whether it is doing what its builders believe.*

This course is methodology for closing that gap — philosophical foundations, technical instruments, and the human capacities that supervise both. The most dangerous failure is not the one that crashes visibly. It is the plausible answer to the wrong question.

WHAT THIS COURSE IS FOR

The cognitive capacities most essential to responsible AI deployment are precisely those where current AI is weakest.

Causal reasoning, metacognitive supervision, counterfactual analysis, and calibrated uncertainty communication. Learning to exercise those capacities rigorously is what this course is for.

TIER 4

Metacognitive & supervisory intelligence

Plausibility auditing, problem formulation, knowing when an output is wrong.

TIER 5

Causal & counterfactual reasoning

Pearl's three-rung ladder — observation, intervention, counterfactual.

TIER 1 — WHAT TOOLS ARE FOR

Pattern recognition & fact retrieval

Delegate freely — and audit relentlessly.

COURSE ARC

From *"should we trust AI?"* to a defended validation pipeline.

PART I

Foundations of Computational Skepticism

Skepticism as method — Descartes, Hume, Popper, Pearl. The Skeptic's Toolkit and the five supervisory capacities.

Weeks 1–3 · Ch 1–3

PART II

Data, Models, Validation

Data validation, explainability, fairness metrics, robustness. The technical infrastructure of doubt.

Weeks 4–7 · Ch 5–8

PART III

Agency & Human–AI Interaction

RL, agentic systems, delegation maps, trust calibration. Where the boundary is drawn — and how it fails.

Weeks 9–10 · Ch 9–10

PART IV

Communication & Cognition

Visualization under validation. Calibrating claims to evidence. The verb taxonomy.

Weeks 11–12 · Ch 11–12

PART V

Ethics, Governance, Limits

Accountability under failure. Philosophical limits the technical infrastructure cannot resolve alone.

Weeks 13–14 · Ch 13–14

PART VI

Independent Research

A complete validation pipeline for a real AI system. Living Deck × 4. Changelog History submitted with the paper.

Week 15 · Final paper

BY THE END OF THIS COURSE, YOU WILL BE ABLE TO

Seven outcomes — every one at Analyze, Evaluate, or Create.

Evaluate

AI outputs for systematic failure modes, distinguishing technical errors from structural misalignment with deployment context.

TIER 4 · METACOGNITIVE

Analyze

Dataset and model bias by tracing it from data collection through model output — and naming the right mitigation.

TIER 4 · METACOGNITIVE

Create

Causal and counterfactual arguments using Pearl's three-rung ladder — observation, intervention, counterfactual.

TIER 5 · CAUSAL

Evaluate

Explainability outputs (SHAP, LIME, counterfactuals) — the gap between what the explanation shows and what domain knowledge requires.

TIER 4 · METACOGNITIVE

Apply

Adversarial attack and defense techniques to demonstrate model fragility under distribution shift.

TIER 4 · METACOGNITIVE

Develop

A research-quality validation pipeline for a real system: methodology, bias audit, explainability, and what the model cannot do.

TIERS 4 & 5

ASSESSMENT ARCHITECTURE

Evidence of *engagement*, not artifact production.

Research project — proposal, midterm, final Project + presentations + paper	~50%
Validation assignments ~8 graded exercises across the term	~20%
Quizzes & exams Criterion-referenced	~10%
Participation & attendance The room only works when it's full	~10%
Quality score Norm-referenced, cohort-relative — how professional review works	20%

Every submission carries a Frictional Journal.

A timestamped **prediction** made before the work begins. A **reflection** completed immediately after. And a required **AI Use Disclosure** ending with one specific thing the model could not do.

The Calibration Baseline is administered three times — non-graded — to track whether you are getting better at knowing what you know, not only at knowing more.

Seven moves that produce evidence of learning when AI can generate the artifact.

1
Predict
Write down what you expect to find — before you start.

2
Lock
Timestamp the prediction. No edits after the fact.

3
Work
Use whatever tools make you better. Including AI.

4
Observe
What did the data, the model, or the user actually do?

5
Reflect
Where did your prediction miss? Where did it land?

6
Trace
What did the AI decide that you hadn't yet decided?

7
Calibrate
Update your priors. Do it visibly, in writing.

The artifact is not evidence of learning by itself. Process, friction, judgment, and transfer are.

SCHEDULE AT A GLANCE

Fifteen weeks, four milestones, one project.

WEEK	PART	TOPIC	MILESTONE
1	I · Foundations	Foundations of AI Skepticism & Botspeak Fluency	Read Ch 1 before class
2	I · Foundations	Logic, Probability, Uncertainty	Assignment 1 — Probability & Uncertainty Audit
3	I · Foundations	Cognitive Bias, Model Bias, Feedback Loops	**Project Proposal due**
4	II · Validation	Data Validation Techniques	Assignment 2 — EDA Skepticism Report
5	II · Validation	Model Explainability & Interpretability	Assignment 3 — Explainability Critique
6	II · Validation	Bias Detection & Mitigation	Assignment 4 — Bias Analysis
7	II · Validation	Robustness & Adversarial Attacks	Assignment 5 — Adversarial Documentation
8	—	Midterm Presentations	**Living Deck v2 — present to class**
9	III · Agency	Reinforcement Learning & AI Agency	—
10	III · Agency	Delegation, Trust, Supervision	Assignment 6 — Delegation Map
11	IV · Communication	Visualization for Transparency	Assignment 7 — Transparency Dashboard
12	IV · Communication	Communicating Findings — peer critique	Living Deck v3 — peer review
13	V · Ethics	Ethics, Governance, Accountability	Assignment 8 — Ethics Case Analysis
14	V · Ethics	Philosophical Limits of AI	Final-project workshop
15	VI · Research	Final Research Presentations	**Final paper · Changelog History**

PART I · WEEKS 1–3

Foundations of *computational skepticism*.

Skepticism as a technical methodology — grounded in Descartes, Hume, Popper, Pearl — and translated into computational practice.

Ch 1
The Skeptic's Toolkit

Ch 2
Probability & Confidence

Ch 3
Bias & Responsibility

Ch 4
The Frictional Method

CHAPTER

01

The Skeptic's Toolkit.

Why should we doubt AI — and what does rigorous doubt actually require?

Four moves you apply *before* trusting any model output: Cartesian doubt, Hume's induction limit, Popperian falsifiability, and the Plato's Cave move.

And the operational form of the book's argument — the *Five Supervisory Capacities* the rest of the course operationalizes as pipeline jobs.

FIVE SUPERVISORY CAPACITIES

- Plausibility auditing — does the output match the world?
- Problem formulation — is this the right question?
- Counterfactual reasoning — what would have happened?
- Calibrated communication — saying what you do and don't know
- Refusal authority — the right to say *do not deploy*

CHAPTER

02

Probability, Uncertainty, and the Confidence Illusion.

What does it mean for an AI system to be "confident" — and why is that different from being correct?

Boolean to probabilistic reasoning.
Frequentist and Bayesian frameworks.
Hume's problem of induction — re-translated as *distribution shift*.

Calibration is the most operationally important diagnostic in the chapter: when high model confidence correlates with high error, the metric is the warning.

FOUR QUESTIONS THAT PREVENT DISASTERS

- What is the base rate?
- What is the conditional you actually need?
- How heavy is the loss tail?
- Is the model calibrated where it matters?

CHAPTER

03

Bias — where it enters, and who is responsible.

Where does bias enter, who is accountable for it, and where is the high-leverage intervention point?

Ten mechanisms, distinguished. Three flavors of bias mapped across them. How bias types compound through a pipeline.

Pearl's Ladder — Rungs 1 and 2 — introduced here, then carried through the rest of the course. The political dimension *is* the technical dimension.

PEARL'S THREE-RUNG LADDER

- Rung 1 · Observation — what correlates with what
- Rung 2 · Intervention — what changes if we act
- Rung 3 · Counterfactual — what would have happened

CHAPTER

04

The Frictional Method.

How do we produce evidence of learning when AI can generate the artifact?

Performance is not learning. The Decoupling Problem — what AI has done to assessment — is not solved by banning the tools.

It is solved by changing what we ask for: a timestamped prediction, the work itself, a reflection, and a Disclosure that ends with one specific thing the AI could not do.

THE SEVEN MOVES

— Predict · Lock · Work

— Observe · Reflect

— Trace · Calibrate

PART II · WEEKS 4–8

Validation surfaces — the *technical infrastructure* of doubt.

Five chapters, five surfaces: data, explanation, fairness, robustness, agency. Each chapter is a lens for asking what the model has actually learned — and what it has not.

Ch 5	Ch 6	Ch 7	Ch 8	Ch 9
Data Validation	Explainability	Fairness	Robustness	Agentic AI

CHAPTER

05

Data Validation — reconstructing the epistemic frame.

Is your dataset actually what you think it is?

Exploratory data analysis as skepticism — not as a procedural pass. EDA reveals the surface; reconstruction reveals the frame.

Why are there exactly N rows? What did the original collector decide for you? What is the access assumption, and where does the boundary really live?

THE INTERROGATION MOVES

- Trace one row to its source
- Name a hidden assumption — defend it or reject it
- Identify the access boundary, not the variable boundary
- Reconstruct the frame the collector worked inside

CHAPTER

06

Model Explainability.

Does an explanation make a model trustworthy — or does it make it appear trustworthy?

SHAP, LIME, counterfactual explanations — each does something specific, and each does *only* that. Explanations describe the model's internal accounting, not the world.

Wittgenstein's language games as a model for interpretability: the same word means different things in the model's game and yours.

THREE WORDS THAT DIFFER

- Explain — describe the model's bookkeeping
- Justify — give a reason that should hold in the world
- Understand — know the system well enough to predict its failures

CHAPTER

07

Fairness Metrics — choosing a definition and defending it.

What counts as fair — and who gets to decide?

The fairness impossibility theorem: three reasonable definitions of fair, one dataset, the arithmetic that proves they cannot all hold. The COMPAS case as worked example.

Each metric is a values claim. The deliverable is not the metric — it is the *defense* of the metric.

THREE DEFINITIONS, ONE IMPOSSIBILITY

- Statistical parity — equal positive rates across groups
- Equalized odds — equal error rates across groups
- Calibration — equal meaning of a score across groups

CHAPTER

08

Robustness — when a pixel breaks the model.

What does it mean for a model to understand something if a perturbation no human would notice can flip the answer?

Adversarial examples expose the difference between human-relevant features and the proxy features the model has actually learned. The panda becomes a gibbon.

The chapter *opens* Pearl's Rung 3 — counterfactuals — that Chapter 13 will close. Honest robustness disclosure is the deliverable.

ROBUSTNESS TOOLKIT — AND ITS COSTS

- Adversarial training — accuracy traded for stability
- Input certification — narrow, expensive, real
- Distribution-shift monitors — detect, do not fix
- Honest disclosure — the limits as part of the spec

CHAPTER

09

Validating Agentic AI.

What changes when the system stops predicting and starts acting?

The categorical shift from prediction to consequence. A taxonomy of agentic failure modes, read against eleven documented cases. Multi-agent patterns that compound the failures.

Ash's case returns here. The agent reported task complete. The task was not complete. The report was fluent, technically accurate, and globally false.

FROM PREDICTION TO ACTION

- Goal misgeneralization — right metric, wrong objective
- Tool misuse — capability without judgment
- Specification gaming — the letter without the spirit
- Compounding errors — small mistakes, irreversible consequences

PART III · WEEKS 10–12

The human side of the human–AI *system*.

Where the boundary between machine and supervisor is drawn — how delegation is contracted, how dashboards argue, how uncertainty survives the trip from your model to your reader.

Ch 10
Delegation & Trust

Ch 11
Visualization

Ch 12
Communicating Uncertainty

CHAPTER

10

Delegation, Trust, and the Supervisory Role.

When should you trust the tool — and what does the override actually look like?

Build a *delegation map* — a contract, not a partition. For each component: what the AI does, what you do, and the testable handoff condition where your domain knowledge overrides the output.

The five supervisory capacities re-appear here as *pipeline jobs*, not personality traits. The Disclosure becomes a supervisory log.

THE HANDOFF CONDITION

- Stated *before* the AI runs
- Testable — falsifiable, not aspirational
- Owned by a named human
- Logged whether triggered or not

CHAPTER

11

Visualization Under Validation.

Does a visualization make us understand — or make us feel like we understand?

A dashboard is an argument. The design choices are normative, not neutral. McLuhan applied to the chart: the medium is the message.

Build the honest version. Then build the deliberately misleading version from the same data. The redesign reveals what the default was already doing.

TWO FACES OF UNCERTAINTY

- Aleatoric — the range the world will produce
- Epistemic — the trust your model has earned
- Show both — or be honest about which you are not showing

CHAPTER

12

Communicating Uncertainty — calibrating claims to evidence.

How do you say what you know and what you don't — without losing the room?

The verb taxonomy: *shows, suggests, demonstrates, proves.* Each verb has an evidentiary requirement. Most engineering writing overstates by one or two verbs.

Reading AI output through the verb taxonomy is one of the highest-leverage supervisory moves available.

THE TAXONOMY, ASCENDING

- Notes — anecdotal, single instance
- Shows — descriptive evidence in this dataset
- Suggests — directional, not yet causal
- Demonstrates — interventional, replicable

PART IV · WEEKS 13–14

Where the technical infrastructure *runs out.*

Two chapters address the territory the toolkit cannot resolve alone — accountability under failure, and the philosophical limits that frame what AI can and cannot mean.

Ch 13
Accountability

Ch 14
Limits of AI

CHAPTER

13

Accountability — who is responsible when the system fails?

When responsibility resists clean attribution, what makes a regime workable?

Two ethics frameworks — Kantian and utilitarian — mapped onto a single topology. The five accountability requirements that any working regime must satisfy.

This chapter *closes* the Pearl Rung 3 question Chapter 8 opened: the governance counterfactual. What would the accountability regime have had to be, for the failure to have been caught?

FIVE REQUIREMENTS

- Pre-specified — written before deployment
- Named — a human, not a department
- Auditable — a record exists, and is read
- Enforceable — there is a consequence
- Revisable — the regime updates on what it learns

CHAPTER

14

The Limits of AI.

What can AI not do — and why does the answer matter for how we build and deploy it?

Three structural limits that capability scaling cannot fix: *meaning, intentionality, the data–world gap.* The Turing Test and the Chinese Room as diagnostic, not decoration.

The book's culminating claim: the supervisor's authority to *refuse deployment* is the most important authority in any human–AI system. The practice this course teaches must include that option — or it is not the practice this course is teaching.

THE THREE STRUCTURAL LIMITS

- Meaning — symbols are not what they refer to
- Intentionality — aboutness is not generated by training
- The data–world gap — the map is not the territory

BY WEEK 15, YOU WILL BE ABLE TO —

Audit any AI deployment and name, for each step, *which capacity is being exercised, and by whom.*

Where the attribution fails, you will have found a gap. Gaps are where patients die in waiting rooms. The course is the practice of finding them — before deployment, at the adoption committee, in the validation review.

Nik Bear Brown, PhD MBA
ni.brown@neu.edu